

Task 1 : TIVIDY Research

We understand a workflow as a model for representing recurring work processes as a structured sequence of tasks that transform an input into an outcome. A Workflow Management System (WfMS) provides the capabilities for modeling, instantiating, executing, and monitoring these workflows. It describes what work must be performed, who or what performs it, in what order it occurs, under what conditions it may continue, and what happens when work succeeds, fails, is rejected, or requires escalation.

Workflow Definitions and Running Instances

SAP's documentation states that, " a workflow definition specifies which actions should be performed, when, and under what circumstances, while a workflow instance is the actual execution of those actions". A single definition can produce multiple independent instances, each with its own data, participants, and progress. Oracle's APEX documentation says, the structure is determined by the Workflow Definition, and every time a workflow starts, a Workflow Instance is created as a specific runtime from that definition. This is essential as it lets us know if the process design is correct and whether a particular workflow instance is progressing correctly.

Representing and Routing Work Items

Work items are the basic units of execution within a workflow. Routing rules are the logic that determines where a work item goes next and who receives it after each step in a workflow. In systems like IBM TRIRIGA, work is organised into hierarchical stages with queues, each representing a phase with its own responsibilities. The key insight is that routing rules must be configurable, not hard-coded, since organisations assign work by different criteria, role, workload, or data-driven rules. Assignment should therefore reference an abstract rule, resolved to participants at runtime, so changing strategy does not require rewriting work-item classes.

Approvals, Rejection, and Escalation

Approval is one of the most common workflow patterns: a work item is sent to an approver who may approve or reject. Microsoft Power Automate's Approvals connector illustrates this, route work items to reviewers, capture decisions, and automatically take the next action. Response types include Approve or Reject. A 2024 patent describes a controller enabling participants to modify their tasks within a flexible framework, deferring task definition to responsible users rather than requiring full specification upfront, supporting dynamic workflows. Escalation handles non-response: Enate describes how external API actions enter a "Waiting" state and, on timeout, escalate to a human queue based on allocation rules.

External Integration and Failure Handling

A WfMS must interact with external systems, and failure handling must be top priority. IBM's webMethods OneData supports external workflow support, passing pending transactions to an external system via iHooks .External approvers can approve, reject, delegate, or save changes. The Enate uses failure handling, timeouts and escalation. TIVIDY should support external integrations, and have flexible ways to handle failures and recover problems.

History, Audit, and Monitoring

The system maintains history to support auditing and monitoring. Microsoft Entra ID Governance records every workflow processing event in audit logs, while Power Automate analyses history to identify bottlenecks and improve performance. HelloID records details such as what and when it happened and who was involved. For TIVIDY, this highlights the importance of maintaining accurate records linked to the specific workflow definition version used, ensuring that past activities can be audited reliably.

Engine Reusability and Definition Changes

A workflow management system consists of different components working together to schedule tasks and handle data. The 2025 terminology paper highlights that the coordination of these components ensures workflow efficiency.Reusability comes from separating configurable decisions from the engine. The 2024 patent stated that users who are assigned to a specific task can define them if they want,which allows the workflow to develop as requirements change.States should be clearly represented in the system so that the WfMS can track the progress and react appropriately to situations.

Key Implications for TIVIDY's Design

Based on the research, TIVIDY should follow several important design principles. 1. workflow definitions must be separated from running instances so that changes to a definition do not unintentionally affect existing instances. 2. we must arrange routing and approval rules while role mappings, approval chains, and escalation thresholds are treated as data. 3.the system must manage the lifecycles of workflow instances and individual activities.It should also support external system integration and provide reliable failure handling. 4. Failure handling must be a priority,external system integration. 5. Auditing and monitoring should be built into the system to ensure that actions, decisions and state changes can be tracked.

Conclusion: This research explored how workflow management systems handle tasks, approvals, routing, failures and monitoring. These findings will guide TIVIDY's design and pattern selection. To make the system reusable, TIVIDY should allow important rules and processes to be configured rather than hard-coded.

References

1. Ferreira da Silva, R., et al. (2025). A terminology for scientific workflow systems. *Future Generation Computer Systems*.
2. SAP. Workflow Definition versus Workflow Instance. SAP Help Portal.
3. Nimblework. (2024). Workflow in Workitem.
4. Microsoft. (2025). Introduction to approval flows. Power Automate Training.
5. US Patent 20240005242A1. (2024). Dynamic federated workflow programming management.
6. IBM. (2025). Using External Workflow in webMethods OneData. IBM Documentation.
7. Microsoft. (2024). Auditing Lifecycle Workflows . Microsoft Entra ID Governance.
8. Enate. (2025). Trigger External API Actions. Enate Help.
9. Tools4ever. (2024). Service Automation – Auditing.
10. Microsoft. (2025). Monitor your flows. Power Automate Guidance.
11. Oracle. (2025). About Workflows. Oracle APEX End User's Guide.

Task 2

Generic Workflow management system for a Construction company

Organisation and Process

We are focusing on a mid to large commercial construction company that manages multiple active building sites at the same time. The specific process we are looking at handles the requisition and financial approval along with the procurement and final site delivery of high value construction materials like custom steel frameworks or bulk concrete.

Participants and Major Work

The main people involved in this workflow include the site manager who starts the process and the procurement officer who reviews the requests. We also have a project finance director for approvals alongside the external supplier and a site quality inspector who receives the goods. The major pieces of work start with drafting the material requisition and sourcing vendor quotes. From there the team issues financial approval before generating the purchase order and handling the physical delivery and quality inspection logging.

Routing Decisions and Rules

The system relies on some important decisions to route work to the right people. If the requisition value is under ten thousand dollars the workflow routes the task directly to the procurement officer for immediate order placement. However if the value is over ten thousand dollars the system routes the request straight to the project finance director for mandatory financial approval before any procurement can happen. If a request sits unapproved for more than 48 hours, an automatic timer triggers an escalation that reassigns it to the project manager.

Scenario: End-to-End Construction Project Lifecycle

Organisation and Process

We are modelling a complete construction project from start to finish for a commercial building company. The process moves naturally from the initial idea through the design and planning phases before hitting the actual construction and the final handover. The main people involved are the client who approves things along the way alongside the lead architect who handles the design. We also have an estimator for finances as well as a project manager to coordinate everything and a site superintendent to run the actual building work.

Major Work and Construction Stages

The main chunks of work begin with the initial proposal and drafting the blueprints. After that the team calculates the budget before moving into the physical construction phase. The construction itself is broken down into several sequential workflow stages. It starts with site preparation and clearing followed by pouring the foundation. Once the foundation sets the team moves on to the structural framing and installing the roof. After the building is sealed they handle the mechanical electrical and plumbing installations. The final building stages cover the interior finishes like drywall and painting before moving on to installing fixtures and completing a final site clean up. Once the building is completely done they conduct a final inspection and sign off the project with the client.

Routing Decisions and Approvals

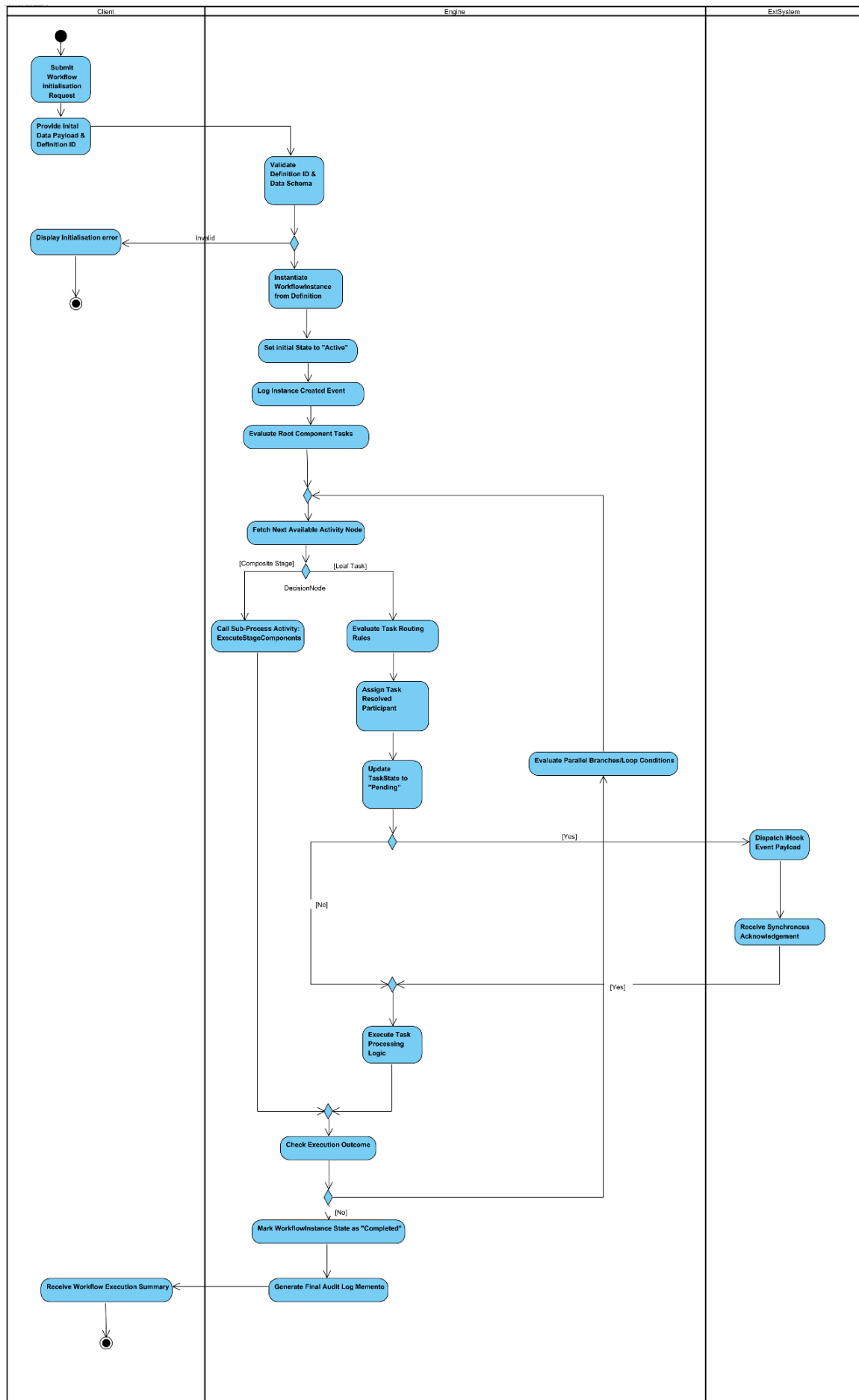
The system makes some key routing decisions throughout these stages. For example if the client rejects the blueprints the task gets sent straight back to the architect for revisions. Also if the final budget is way higher than expected the workflow forces the project manager to reduce the scope before any actual site preparation can begin.

Dependencies Failures and Outcomes

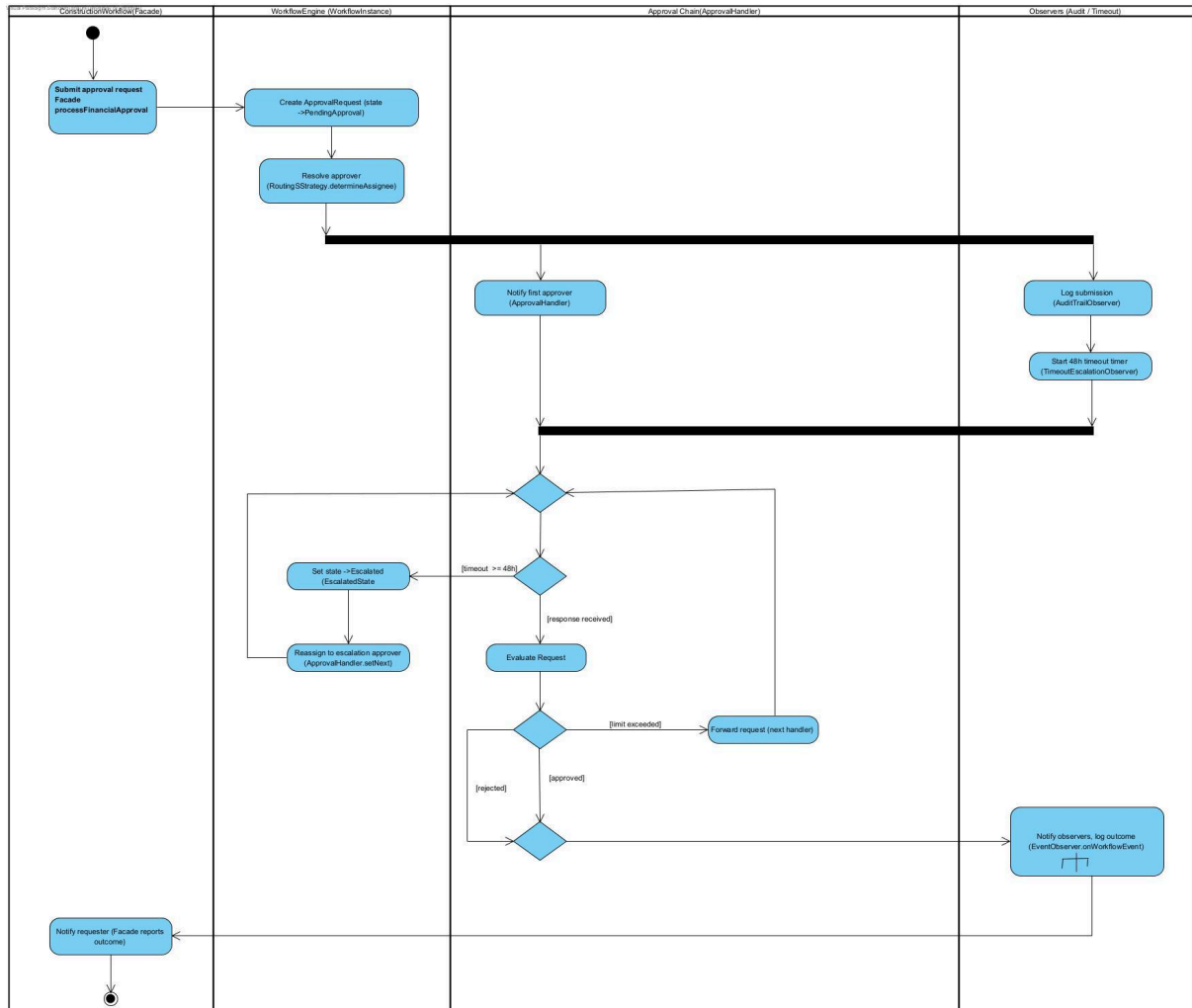
The system relies on strict dependencies to keep things in order. Actual construction cannot start until the blueprints and the budget are fully approved. Moving from one building stage to the next also requires the previous stage to be fully completed. Handing the building over to the client depends entirely on passing the final inspection. If something goes wrong like a municipal permit taking too long the system automatically escalates the issue to the project manager. If a specific construction stage fails an inspection the workflow loops back to a defect resolution phase for that exact stage. We can also integrate external systems to handle automatic permit submissions and accounting software for generating invoices. The final expected outcome is a fully completed building with client approval and all invoices sent out successfully. The system also connects to external systems for municipal permits and email alerts, while logging an activity history for every approval and rejection.

Task 3: UML Activity Diagrams

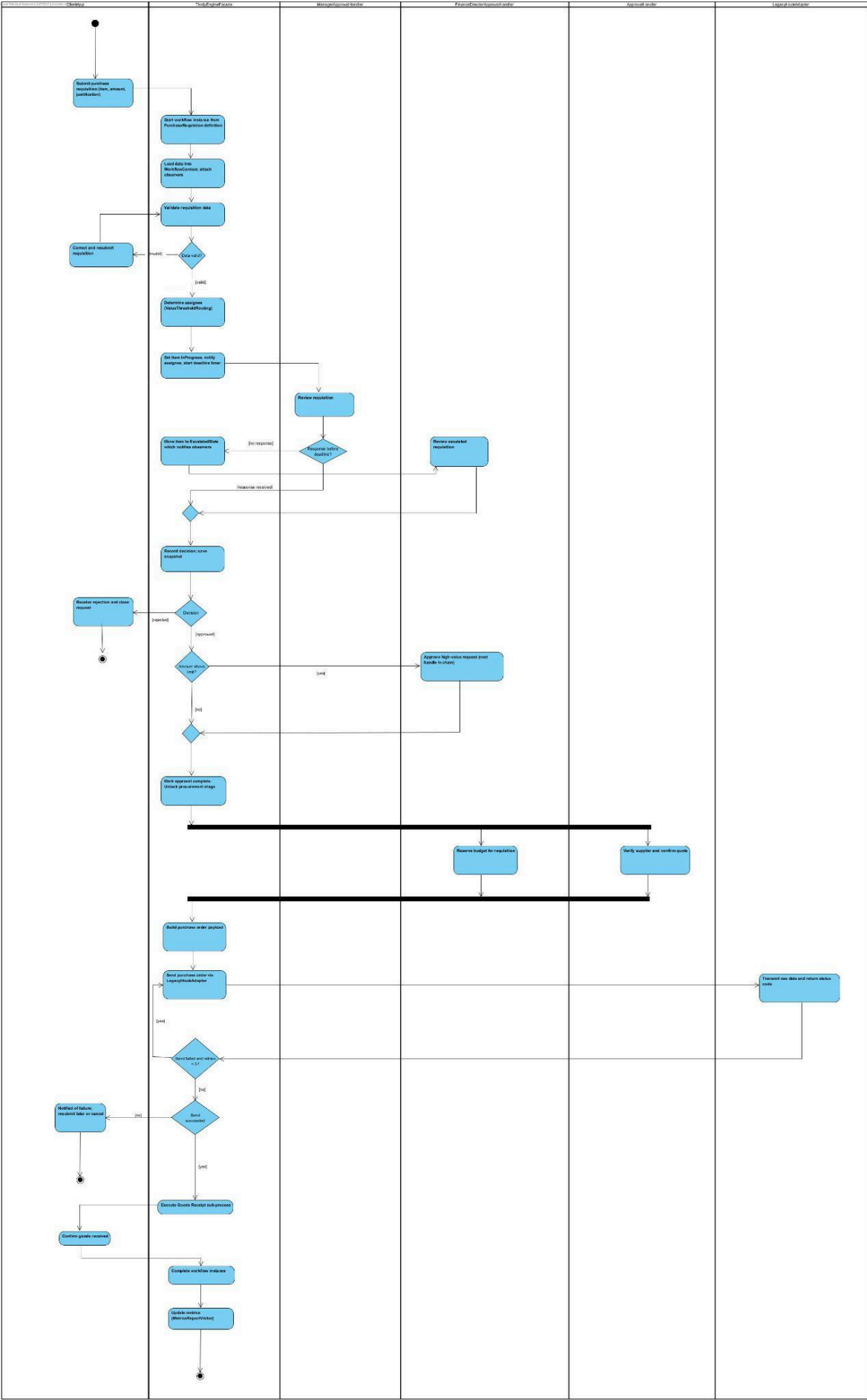
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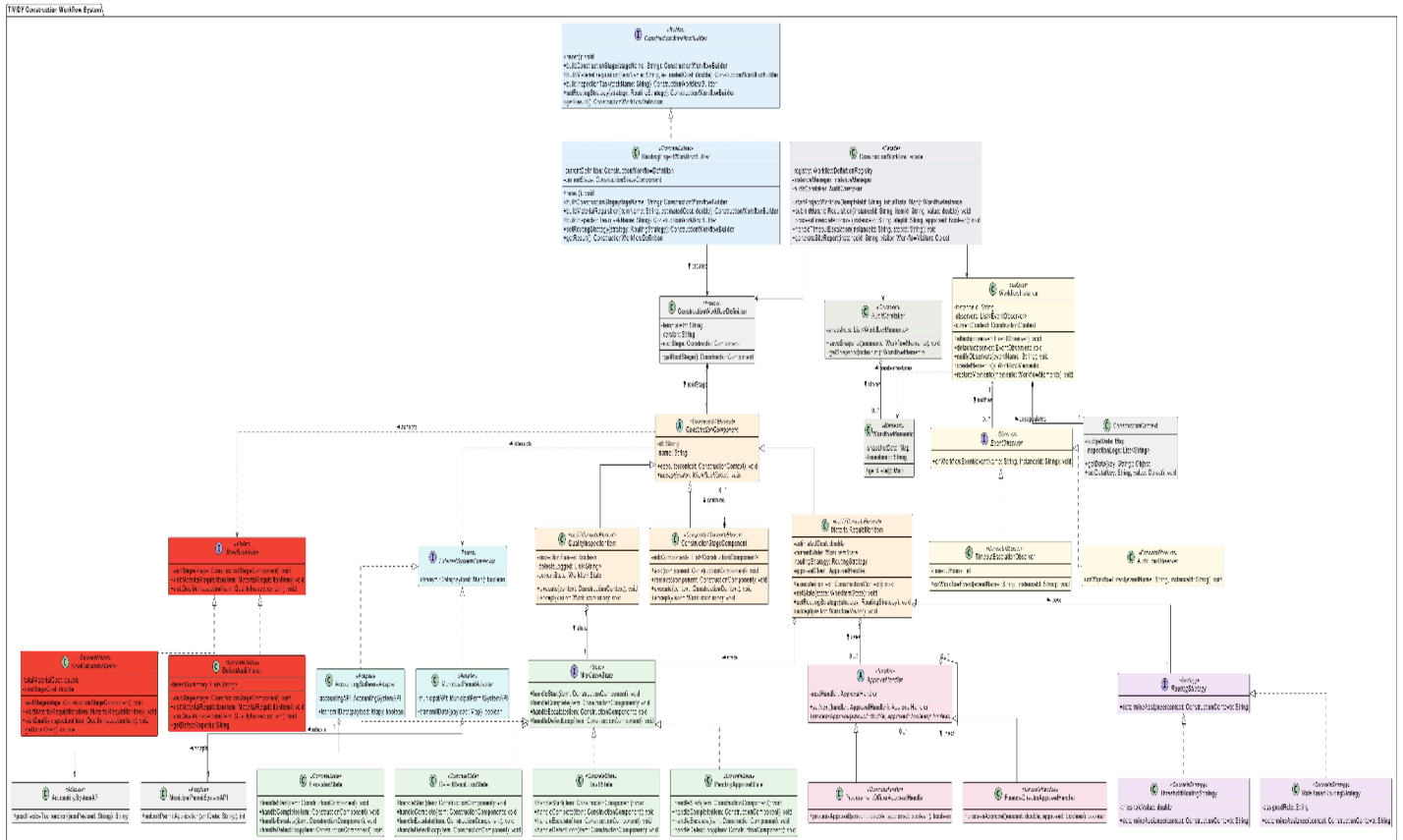


Task 4: GoF Design Patterns

Design Patterns	Participants
Builder	ConstructionWorkflowBuilder, BuildingProjectWorkflowBuilder
Observer	WorkflowInstance, EventObserver, TimeoutEscalationObserver, AuditTrailObserver
Strategy	RoutingStrategy, ThresholdRoutingStrategy, RoleBasedRoutingStrategy
Chain Of Responsibility	ApprovalHandler, ProcurementOfficeApprovalHandle, FinancialDirectorApprovalHandler
State	WorkItemState, DraftState, PendingApprovalState, DefectResolutionState, EscalatedState
Visitor	WorkflowVisitor, CostCalculatorVisitor, DefectAuditVisitor
Adapter	ExternalSystemConnectorAccountingSoftwareAdapter, MunicipalPermitAdapter, MunicipalPermitSystemAPI, AccountingSystemAPI
Composite	ConstructionWorkflowDefinition, ConstructionComponent, QualityInspectionItem, ConstructionStageComponent, MaterialRequisitionItem
Facade	ConstructionWorkflowFacade
Memento	ConstructionWorkflowFacade, AuditCaretaker, WorkflowMemento,

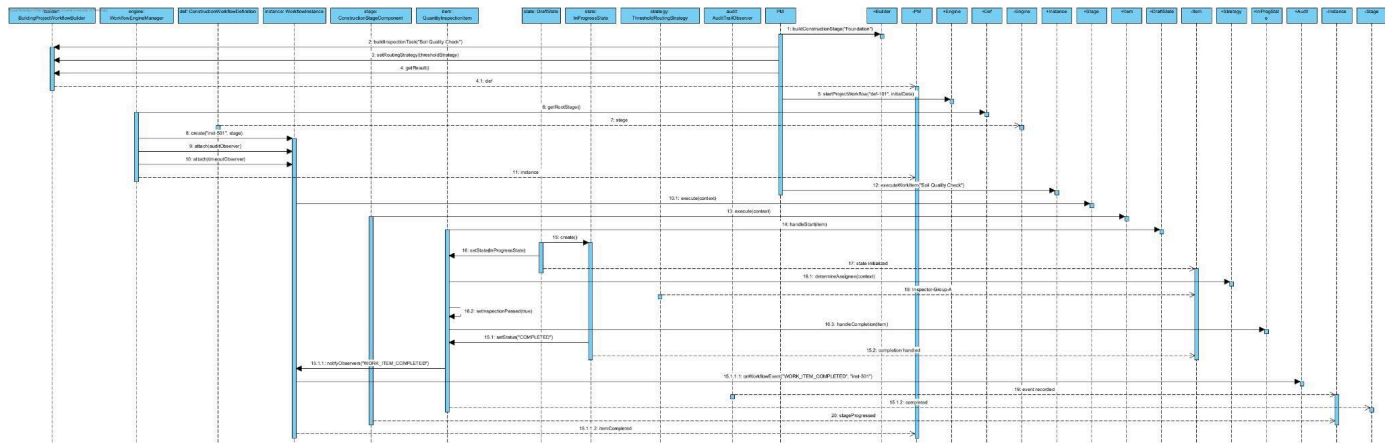
Task 5: Class Diagram

Builder (Constructive Process Control)
Compelling Evidence: Bridges & Walton (Principles)
Strategic Process Design (Detailed)
Strategy #1 (The Theistic & Role Redefinition)
Chain of Responsibility (Financial & Knowledge & Resources)
Resource Site Allocation (Detailed)
Adaptive Planning: Permit to Self-Organize Software
Management Unit & Structure & Culture
Process Site Control & Project & Methodologies
Project Control: The Managerial Engineering

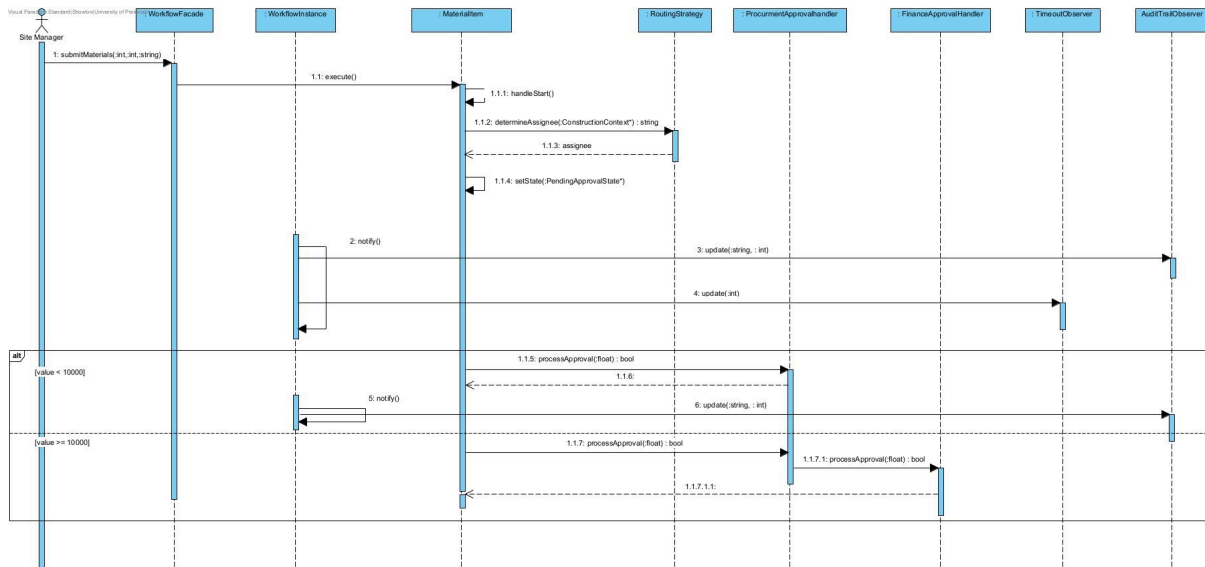


Task 6: Runtime Behaviour Diagrams

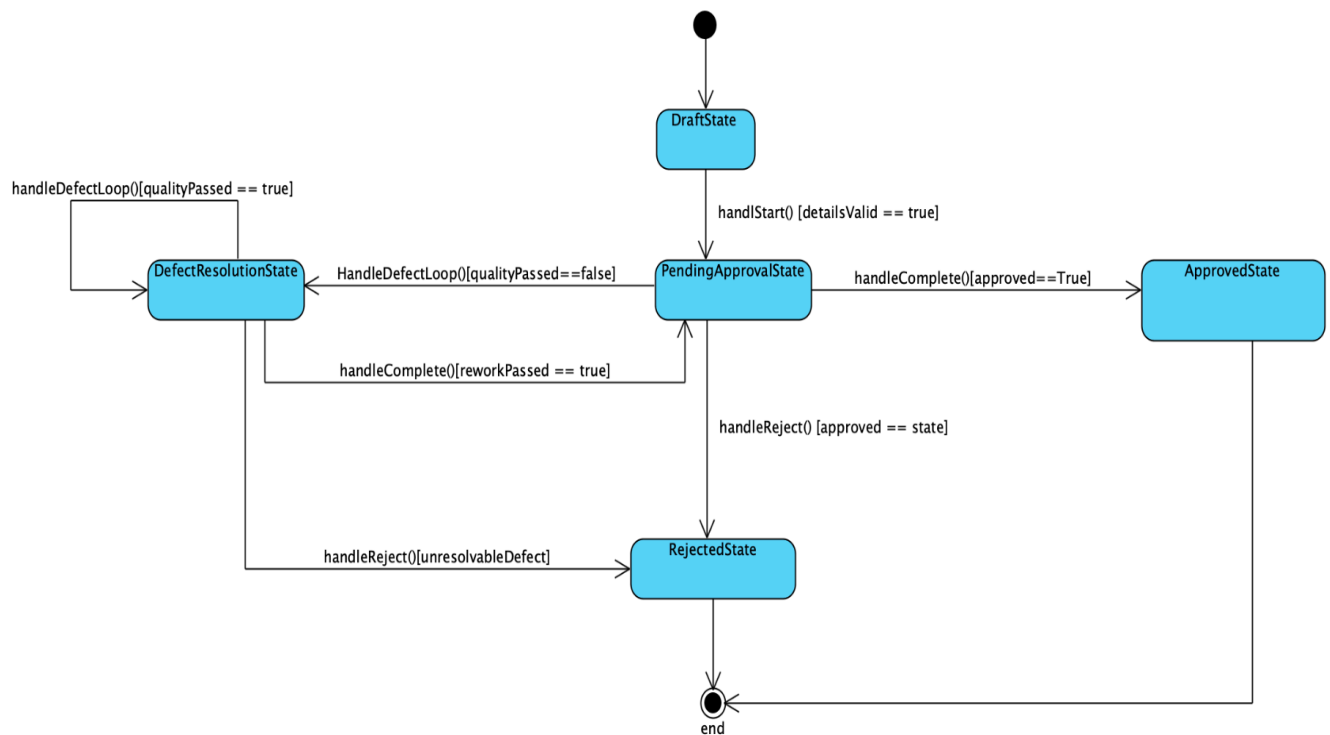
1. Sequence Diagram 1



2. Sequence Diagram 2



3. State diagram



Team Details

1. Lisakhanya Tatane u25514424
2. Tendai Jonasi u25010230
3. Siphwe Aphane u24876829
4. Lethukuthula Ngwenya u24986535
5. Munashe Chitoro u24888096
6. Showlon Hunter u25502469

Github Repository Link

<https://github.com/user6829/COS214-Prac6>